Neonatal Point-of-Care Ultrasound Course

Preliminary Programme — 31 May to 3 June 2027

Preliminary — subject to change

Course overview

The course runs over four days for 60 participants and covers four topics: cardiac, lung, GI/abdominal and vascular access ultrasound. Days 1–3 are the lecture programme; Day 4 is entirely hands-on.

Day	Date	Main focus
Day 1	Monday 31 May	Cardiology and haemodynamics, and GI ultrasound (incl. NEC)
Day 2	Tuesday 1 June	Lung ultrasound, vascular access, GI ultrasound and ultrasound-guided lumbar puncture; PDA assessment and management (online)
Day 3	Wednesday 2 June	Vascular access, GI ultrasound, the crashing baby protocol and neuro (sinus venous thrombosis), haemodynamics in asphyxia; elective workshops
Day 4	Thursday 3 June	Hands-on scanning and small-group discussions

Day 1 — Monday 31 May 2027

Cardiology, Haemodynamics and GI Ultrasound

Time	Area	Topic	Duration
08:00–08:30	—	*Registration and welcome coffee*	*30 min*
08:30–08:45	—	*Welcome, course aims and structure*	*15 min*
08:45–09:25	General	Ultrasound Physics and Knobology: Optimising the Neonatal Image	40 min
09:25–09:45	—	*Coffee break*	*20 min*
09:45–10:30	Cardiac	Neonatal Echocardiography I: Standard Views and Normal Anatomy	45 min
10:30–11:10	GI	Introduction to Abdominal POCUS: Systematic Approach and Normal Anatomy	40 min
11:10–11:50	Cardiac	Neonatal Echocardiography II: Assessing Cardiac Function and Filling	40 min
11:50–12:35	GI	Bowel Ultrasound and Necrotising Enterocolitis (NEC) — Prof. Alan Daneman, SickKids, Toronto	45 min
12:35–13:35	—	*Lunch break*	*60 min*
13:35–14:05	Cardiac	Tamponade and Pericardial Effusion	30 min
14:05–14:50	GI	Malrotation, Volvulus and Other Surgical Abdominal Emergencies	45 min
14:50–15:10	—	*Coffee break*	*20 min*
15:10–15:55	Cardiac	Pulmonary Hypertension and Right Ventricular Function	45 min
15:55–16:30	—	*Questions, discussion and close of Day 1*	*35 min*

Day 2 — Tuesday 1 June 2027

Lung Ultrasound, Vascular Access, GI and Lumbar Puncture, with the PDA

Time	Area	Topic	Duration
08:00–08:20	—	*Welcome to Day 2 and recap of Day 1*	*20 min*
08:20–09:10	Lung	ABC of Lung Ultrasound: Patterns and Findings	50 min
09:10–09:50	Vascular access	Introduction to Ultrasound-Guided Vascular Access: Vascular Anatomy and Vessel Assessment	40 min
09:50–10:10	—	*Coffee break*	*20 min*
10:10–10:55	Vascular access	Ultrasound-Guided Insertion Techniques and Needle Guidance	45 min
10:55–11:40	GI	Renal and Urinary Tract Ultrasound	45 min
11:40–12:20	Lung	LUS in the Neonatal Unit	40 min
12:20–13:20	—	*Lunch break*	*60 min*
13:20–14:00	Procedures	Ultrasound-Guided Lumbar Puncture: Spinal Anatomy, Landmarks and Needle Guidance	40 min
14:00–14:40	Vascular access	Visualisation of Superficial Veins and Epicutaneo-Caval Catheter (ECC) Insertion	40 min
14:40–15:10	—	*Coffee break (video link check for the online lectures)*	*30 min*
15:10–15:50	Cardiac	The PDA: Echocardiographic Assessment of Haemodynamic Significance (live online)	40 min
15:50–16:30	Cardiac	PDA Management: Pharmacological and Transcatheter Closure, with questions and close of Day 2 (live online)	40 min

Day 3 — Wednesday 2 June 2027

Vascular Access, GI, the Crashing Baby, Neuro, Asphyxia and Elective Workshops

Time	Area	Topic	Duration
08:00–08:15	—	*Recap of Day 2*	*15 min*
08:15–08:55	Vascular access	Catheter Tip Localisation and Catheter-Related Complications: UVC, UAC and PICC; Malposition, Thrombosis and Effusions	40 min
08:55–09:40	GI	Liver and Adrenals: Acute Abdominal Bleeds, incl. Adrenal Haemorrhage	45 min
09:40–10:00	—	*Coffee break*	*20 min*
10:00–10:40	Vascular access	Prevention of CLABSI: Securement and Exit-Site Protection	40 min
10:40–11:20	Emergency	The Crashing Baby Protocol	40 min
11:20–11:55	Neuro	Neuro: Sinus Venous Thrombosis	35 min
11:55–12:00	—	*Questions and discussion*	*5 min*
12:00–13:00	—	*Lunch break*	*60 min*
13:00–13:30	Cardiac	Haemodynamics in Asphyxia	30 min
13:30–13:40	—	*Introduction to the elective workshops and participant allocation*	*10 min*
13:40–14:30	Elective	Elective Workshop — Part 1	50 min
14:30–14:50	—	*Coffee break*	*20 min*
14:50–15:40	Elective	Elective Workshop — Part 2	50 min
15:40–16:30	Elective	Elective Workshop — Part 3	50 min
16:30–17:00	—	*Workshop feedback, discussion and preparation for Day 4*	*30 min*

Elective workshops

Each participant stays in one workshop for all three parts. The workshops are informal, built around small-group discussion and clinical cases. Depending on the topic, they may include simple scanning exercises on fellow participants and EchoPAC measurement training.

- Cardiac — neonatal echocardiography and haemodynamics
- Lung — lung ultrasound
- GI/Abdominal — abdominal POCUS
- Vascular access — ultrasound-guided vascular access

Day 4 — Thursday 3 June 2027

Hands-On Workshops

08:00–16:15 at the Clinical Training Centre (KTC), Queen Silvia Children's Hospital, Östra sjukhuset. Hands-on scanning and small-group discussions across all four topics, with all 60 participants working at the same time in 10 groups of 6.

Training takes place on:

- healthy children and infants (volunteers)
- simulators
- stable babies in the NICU, depending on availability on the day

Faculty

- Prof. Yogen Singh, UC Davis, Sacramento, USA — haemodynamics
- Prof. Alan Daneman, SickKids, Toronto, Canada — GI and NEC ultrasound
- Fiammetta Piersigilli, Brussels, Belgium — vascular access
- Nadya Yousef, Paris, France — lung ultrasound
- John Madar, Plymouth, UK — haemodynamics
- Instructors from Gothenburg, Stockholm, Halmstad and Borås